Eric A. Chadwick

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LinkedIn: Eric A. Chadwick
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Education

May., 2020 – Jun., 2025 — **PhD, University of Toronto** Mechanical & Industrial Engineering
Thesis title: *Microscale Transport in Polymer Electrolyte Membrane Fuel Cells: Impact of Gravity and Flow Field Design.*

May, 2017 – Sep., 2019 — **MASc, University of Toronto** Mechanical & Industrial Engineering
Thesis title: *Vessel Network Extraction via micro-CT Imaging: A Foundation for Modelling Lung De- and Recellularization*

Sep., 2012 – Apr., 2017 — **BEng, University of Guelph** Mechanical Engineering
Thesis title: *Smart Thermal Battery System*

Employment History

Nov., 2019 – Present — **Academic Staff & Researcher**, Baden-Württemberg Cooperative State University
Courses Taught: *Numerical Methods for Engineers, Introduction to Python, Transport in Porous Media*

May, 2020 – Jan., 2025 — **Research Assistant (PhD)**, University of Toronto

Sept. – Oct., 2020 — **Lab Demonstrator Teaching Assistant**, University of Toronto
Course: *Mechanical & Thermal Energy Conversion*

Jan. – Apr., 2018 — **Head Tutorial Teaching Assistant**, University of Toronto
Course: *Heat & Mass Transport*

May, 2017 – Sep., 2019 — **Research Assistant (MASc)**, University of Toronto

Jan., 2015 – Aug., 2016 — **Undergraduate Research Assistant**, University of Guelph
Research Topic: *Multiphase flow and efficiency analysis of an air lift pump*

May, 2015 – Aug., 2015 — **Quality Engineering Intern**, Ontario Power Generation

May, 2014 – Apr., 2015 — **Undergraduate Research Assistant**, University of Guelph
Research Topic: *Minimum quantity lubrication strategies for the machining of hard-to-cut materials*

Research Publications

Journal Articles

1. **E.A. Chadwick**, B. Derebaşı, V. P. Schulz, and A. Bazylak, "Influence of gravity on water management in polymer electrolyte membrane fuel cells," *Scientific Reports, Special Issue: Transport Phenomena in Electrochemical Devices*, Accepted.

2. Y. Dumay, **E.A. Chadwick**, M. Napoléon, C. Seide, B. Ortlepp, N. Blessing, and V. P. Schulz, "Development of an additively manufactured vapor chamber using hybrid wick structures," *Applied Thermal Engineering*, p. 126 442, 2025. URL: `https://doi.org/10.1016/j.applthermaleng.2025.126442`.

3. **E.A. Chadwick**, P. Shrestha, H. B. Parmar, A. Bazylak, and V. P. Schulz, "Biomimetic auxiliary channels enhance oxygen delivery and water removal in polymer electrolyte membrane fuel cells," *Applied Energy*, vol. 389, 2025. URL: `https://doi.org/10.1016/j.apenergy.2025.125760`.

4. D. Parekh, S. Ranieri, T. Seip, **E.A. Chadwick**, B. Derebaşı, N. Ge, R. Wang, and A. Bazylak, "Dominating impact of microporous layer thickness on gas diffusion layer oxygen transport resistance," *Journal of Power Sources*, vol. 656, p. 238 031, 2025. URL: `https://doi.org/10.1016/j.jpowsour.2025.238031`.

5. J. K. D. Chan, **E.A. Chadwick**, T. Taniguchi, A. Ahmadipour, T. Suzuki, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, "Predicting cell distributions in lung vasculature to optimize re-endothelialization," *Frontiers in Bioengineering and Biotechnology*, vol. 10, pp. 2296–418, 2022. URL: `https://10.3389/fbioe.2022.891407`.

6. **E.A. Chadwick**, L. K. Hammen, V. P. Schulz, A. Bazylak, M. A. Ioannidis, and J. T. Gostick, "Incorporating the effect of gravity into image-based drainage simulations on volumetric images of porous media," *Water Resources Research*, vol. 58, e2021WR031509, 2022. URL: `https://doi.org/10.1029/2021WR031509`.

7. **E.A. Chadwick**, T. Suzuki, M. G. George, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, "Vessel network extraction and analysis of mouse pulmonary vasculature via x-ray micro-computed tomographic imaging," *PLOS One Computational Biology*, vol. 17(4), e1008930, 2021. URL: `https://doi.org/10.1371/journal.pcbi.1008930`.

Conference Proceedings

1. R. Abed, **E.A. Chadwick**, and W. H. Ahmed, "Two-phase flow behaviour in airlift pumps," in *Proceedings of the 5th International Conference of Fluid Flow, Heat and Mass Transfer*, Niagara Falls, Canada, 2018, pp. 7–9. URL: `https://avestia.com/FFHMT2018_Proceedings/files/paper/FFHMT_168.pdf`.

Conference Presentations

1. **E.A. Chadwick**, A. Briand, T. Seip, B. Auvity, A. Bazylak, and V. P. Schulz, *Balancing performance and pressure drop in Ni foam flow fields via embedded channel design*, Presented at PRiME 2024, Honolulu, USA, 2024. URL: `https://iopscience.iop.org/article/10.1149/MA2024-02443086mtgabs/meta`.

2. **E.A. Chadwick**, V. P. Schulz, and A. Bazylak, *Effects of gravity on water transport in PEM fuel cells via synchrotron x-ray radiography*, Presented at PRiME 2024, Honolulu, USA, 2024. URL: `https://iopscience.iop.org/article/10.1149/MA2024-02443076mtgabs/meta`.

3. **E.A. Chadwick**, P. Shrestha, H. B. Parmar, V. P. Schulz, and A. Bazylak, (General Student Poster Award Winner, 3rd Place) *biomimetic auxiliary channels for enhanced pem fuel cell performance*, Presented at PRiME 2024, Honolulu, USA, 2024. URL: `https://iopscience.iop.org/article/10.1149/MA2024-02674661mtgabs/meta`.

4. T. Seip, J. Lee, L. Zhu, S. Ranieri, **E.A. Chadwick**, K. Krause, L. Helfen, B. Auvity, S. Chevalier, and A. Bazylak, *Elucidating 3D water distributions in PEM water electrolyzers via neutron imaging*, Presented at PRiME 2024, Honolulu, USA, 2024. URL: `https://iopscience.iop.org/article/10.1149/MA2024-02453103mtgabs/meta`.

5. L. Zhu, J. Lee, C. T. Ham, **E.A. Chadwick**, H. B. Parmar, P. Tritk, and A. Bazylak, *Compression of porous transport layers enhanced bubble transport in polymer electrolyte membrane water electrolyzers*, Presented at PRiME 2024, Honolulu, USA, 2024. URL: `https://iopscience.iop.org/article/10.1149/MA2024-02453141mtgabs/meta`.

6. **E.A. Chadwick**, H. B. Parmar, P. Shrestha, A. Bazylak, and V. P. Schulz, *Improving water management and reactant distribution in PEM fuel cells via flow fields with biomimetic auxiliary channels*, Presented at ECS 244, Gothenburg, Sweden, 2023. URL: `https://iopscience.iop.org/article/10.1149/MA2023-02381859mtgabs/meta`.

7. H. B. Parmar, **E.A. Chadwick**, P. Shrestha, and A. Bazylak, *Tailoring parallel channel flow fields for efficient mass transport and compression in proton exchange membrane fuel cells*, Presented at ECS 244, Gothenburg, Sweden, 2023. URL: `https://iopscience.iop.org/article/10.1149/MA2023-02381861mtgabs/meta`.
8. **E.A. Chadwick**, H. B. Parmar, P. Shrestha, A. Bazylak, and V. P. Schulz, *Biomimetic microchannels for the passive management of water in pem fuel cells*, Presented at ECS 242, Atlanta, USA, 2022. URL: `https://iopscience.iop.org/article/10.1149/MA2022-02401455mtgabs/meta`.
9. H. B. Parmar, **E.A. Chadwick**, P. Shrestha, and A. Bazylak, *Tailoring flow field channel aspect ratio for efficient mass transport and compression in fuel cells*, Presented at ECS 242, Atlanta, USA, 2022. URL: `https://iopscience.iop.org/article/10.1149/MA2022-02401452mtgabs/meta`.
10. **E.A. Chadwick**, L. K. Hammen, M. A. Ioannidis, V. P. Schulz, J. T. Gostick, and A. Bazylak, *Elucidating the effects of gravity when simulating drainage in porous media via an image-based porosimetry method*, AGU Fall Meeting Abstracts, 2021. URL: `https://ui.adsabs.harvard.edu/abs/2021AGUFM.H22D..05C/abstract`.
11. J. K. D. Chan, **E.A. Chadwick**, T. Taniguchi, A. Ahmadipour, T. Suzuki, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, *Modelling of flow-induced shear stress to predict targeted delivery of cells in the decellularized lung*, Proceedings of The Joint Canadian Society for Mechanical Engineering and CFD Society of Canada International Congress, Abstract ID: 141, 2019.
12. **E.A. Chadwick**, T. Suzuki, M. G. George, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, *Multiscale fluid flow analysis of decellularized mouse lungs to improve lung regeneration*, Proceedings of The Joint Canadian Society for Mechanical Engineering and CFD Society of Canada International Congress, Abstract ID: 203, 2019.
13. **E.A. Chadwick**, T. Suzuki, M. G. George, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, *In-silica extraction of mouse lung vasculature via micro-CT for modelling de- and recellularization*, Presented at Biomedical Engineering Society Annual Meeting, Atlanta, USA, Abstract ID: 718, 2018.
14. **E.A. Chadwick**, T. Suzuki, M. G. George, D. Romero, C. Amon, T. K. Waddell, K. G, and A. Bazylak, *Micro-CT study of mouse lung vasculature: A foundation for modelling lung de- and recellularization*, Presented at Biomedical Engineering Society Annual Meeting, Phoenix, USA, Abstract ID: 1382, 2017.

Skills

Languages	English (Native), German (A2)
Coding	Python, Matlab, C++, ImageJ Macro Language
Software	Fiji (ImageJ), SolidWorks, AutoDesk CFD, AutoDesk Inventor, GeoDict, Dragonfly, COMSOL, OriginPro, Inkscape, Microsoft Office
Misc.	Academic research, teaching, training, mentoring

Awards & Scholarships

2024 – 2025	**Ontario Graduate Scholarship**, University of Toronto ($10,000 CAD)
	Hatch Graduate Scholarship for Sustainable Energy Research, University of Toronto ($10,000 CAD)
2023 – 2024	**Queen Elizabeth II Graduate Scholarship in Science & Technology**, University of Toronto ($15,000 CAD)
2023	**Climate Positive Energy Scholarship**, University of Toronto ($15,000 CAD)
	Mitacs Globalink Research Award, Mitacs ($6,000 CAD)

Awards & Scholarships (continued)

- 2023 – 2024 **Hatch Graduate Scholarship for Sustainable Energy Research**, University of Toronto ($10,000 CAD)
- 2022 **Faculty of Applied Science & Engineering Graduate Student Endowment Fund**, University of Toronto ($6,250 CAD)
- **Hatch Graduate Scholarship for Sustainable Energy Research**, University of Toronto ($10,000 CAD)
- 2021 – 2022 **Queen Elizabeth II Graduate Scholarship in Science & Technology**, University of Toronto ($15,000 CAD)
- **Ontario Graduate Scholarship**, University of Toronto ($10,000 CAD - Declined due to overlap with Queen Elizabeth Scholarship)
- 2020 – 2021 **Ontario Graduate Scholarship**, University of Toronto ($10,000 CAD)
- 2018 – 2019 **Ontario Graduate Scholarship**, University of Toronto ($10,000 CAD)
- 2017 **Barbara & Frank Milligan Graduate Fellowship**, University of Toronto ($5,460 CAD)
- 2016 **Undergraduate Research Assistant Award**, University of Guelph ($6,989 CAD)
- 2014 **Undergraduate Research Assistant Award**, University of Guelph ($6,989 CAD)
- 2013 **Lindsay W. Christie Bursary**, University of Guelph ($1,000 CAD)

Certifications

- 2019 **Certificate of Data Science**, SciNet
  Courses: *Neural Networking Programming, Quantitative Applications for Data Analysis*
- **Second Degree Black Belt**, Gōjū-ryū Karate
- 2011 **First Degree Black Belt**, Gōjū-ryū Karate